Jiarui He (何佳芮)

Department of Mechanical Engineering, Zhejiang University

Tactile Perception | Underwater Robotics | Embodied AI

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Education

- **Master: Zhejiang University**, Hangzhou, China .... 09/2024 – Present
  Laboratory: The State Key Laboratory of Mechatronic Systems, Department of Mechanical Engineering
  Advisors: Prof. Peng Zhao, Department of Mechanical Engineering
- **Bachelor: Zhejiang University**, Hangzhou, China .... 09/2020 – 06/2024
  Major: Mechanical Engineering
  GPA: Overall: 3.91/4

Publications

- **Jiarui He**, Huangzhe Dai, Kan Liu, Neng Xia, Chengfeng Pan, Chengqian Zhang, and Peng Zhao, "Biomimetic magnetic cilia sensor for flow vector measurement and motion state estimation of aquatic objects", submitted to ***IEEE Transactions on Instrumentation and Measurement(TIM)***, Under Review, May 2026.
- **Jiarui He**, Yan Zhou, Chengqian Zhang, Huangzhe Dai, Chengfeng Pan, and Peng Zhao, "Online Velocity Estimation of a Robotic Fish Using Artificial Lateral Line System With Velocity-Decoupling Sensing Ability", accepted by ***IEEE Robotics and Automation Letters(RA-L)***, vol. 10, no. 10, pp. 10418–10425, Oct. 2025.
  [Publication] [Preprint_PDF]

Extracurricular Working & Study Experience

- Incoming **Research Intern**, **Sharpa Robotics**, Tactile Department, Shanghai, China 06/2026 – 09/2026
- Participated in the **Singapore Summer Academic Exchange Program 2023**. Visited the National University of Singapore (NUS), Nanyang Technological University (NTU), A*STAR, and SIMTech for academic exchange. 08/2023
- Participated in a summer industry internship program at **DEEP Robotics**, Hangzhou, China, with exposure to quadruped robot hardware, locomotion control, and real-world robotic system deployment. 07/2023
- Participated in an online exchange program at **the University of Hong Kong**, focusing on frontier topics in robotics. 08/2021

Projects & Research Experience

- **Biomimetic Soft Robotic Fish with Distributed Flow Field Sensing Capability** 09/2023 – 07/2024
  Independent Research, *Supervised by Prof. Tiefeng Li, Zhejiang University*
  - Designed and fabricated a biomimetic manta ray-inspired robotic fish with velocity sensing capability, including mechanical and hardware design, circuit integration and embedded control programming.
  - The work was presented as a poster at the 5th China Robotics Academic Annual Conference.
- **Infrared Gas Annotation and Detection** 07/2022 – 06/2023
  Independent Research, *Supervised by Prof. Jin Wang, Zhejiang University*
  - Built convolutional neural network (CNN) architectures, including **YOLO** and **PicoDet** models.
  - Annotated infrared gas imagery from large-scale scene datasets for model training and evaluation.
- **Design and Manufacturing of a Cargo Transportation Drone** 07/2022 – 09/2022
  Group Leader, *National Demonstration Center for Experimental Engineering Education, Zhejiang University*
  - Completed the structural design of the mechanical components of a drone, and assembled the prototype using partially 3D-printed parts.
  - Implemented the motion control of the drone and accomplished cargo transportation tasks.

Leadership & Service

- **Reviewer**: RA-L (2025)
- **Teaching assitant**: Engineering Materials, Zhejiang University, 2024 Fall
- Served as **President of the Graduate Student Association**, Department of Mechanical Engineering, Zhejiang University; served as **Chief Director** of the department's New Year Gala, attracting over 1,000 attendees. 09/2024 – 09/2025
- **Volunteer**: "Five-Star Volunteer" award, Zhejiang University, with over **400** cumulative volunteer hours; led in two month-long volunteer teaching programs

Honors & Awards

Graduate Level:

- **National Scholarship** (**Top scholarship in China, 0.2% domestically**) 10/2025
- **Top Ten Students of the Year**, Department of Mechanical Engineering, Zhejiang University (**Top 10 of approximately 1,500 students**). 2024 – 2025
- Second Prize, Zhejiang University Division of the 7th China Graduate Robotics Innovation Design Competition (Ranked **1st** among Second Prize teams and 3rd overall out of 13 teams) 06/2025
- **Outstanding Graduate Student**, Department of Mechanical Engineering, Zhejiang University 2024 – 2025
- **Five-Excellence Graduate Student**, Department of Mechanical Engineering, Zhejiang University 2024 – 2025
- **Outstanding Graduate Student**, Department of Mechanical Engineering, Zhejiang University 2024 – 2025

Undergraduate Level:

- **Outstanding Undergraduate Graduate**, Zhejiang University 2024
- Second Prize, Regional Competition of the "Jingdiao Cup", China College Students Mechanical Engineering Innovation and Creativity Competition (**Top 1%**) 06/2024
- **Best Presentation Award** of the Undergraduate Thesis Project, Department of Mechanical Engineering, Zhejiang University (**awarded to 10 out of 200 students**) 06/2024
- Second-Class Scholarship, Zhejiang University 10/2023
- Hengli Second-Class Scholarship 10/2023
- **National Third Prize**, 16th National College Student Social Practice and Science Contest on Energy Saving and Emission Reduction; advanced from Zhejiang University's campus-level competition with **University First Prize**. 06/2023
- Second Prize, China Undergraduate Mathematical Contest 04/2023
- Second Prize, 9th ZWSoft Cup Engineering Practice and Innovation Competition, Zhejiang university 11/2022
- Zhejiang Provincial Government Scholarship (**Top 5%**) 10/2022
- Second-Class Scholarship, Zhejiang University 10/2022
- Third-Class Scholarship, Zhejiang University 10/2021

Skills & Interests

Programming: Python/Pytorch, C/C++, MATLAB, ROS2, Embedded (STM32), Arduino...
Mechanical Design: SolidWorks, AutoCAD, Abaqus, Ansys...
Fabrication: 3D printing (FDM/DLP), Soft material fabrication...
Language: TOFEL: 111 (R30 L29 S25 W27)
Interests: Photography, Music (Piano & Vocal, Both Grade 10), Tennis, Travelling...